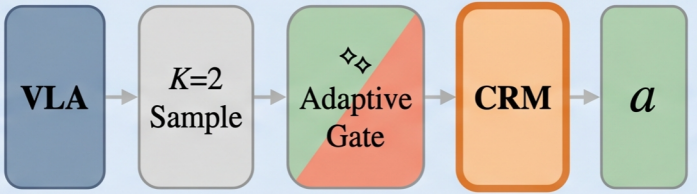
Frozen VLA at Test Time



- Fixed weights (0.4-3.6B params)
- No task-specific fine-tuning
- Variable action quality ⚠

Adaptive Conformal Refinement



~327K params | 1.3 MB | ~170ms latency

Adapts intervention to VLA uncertainty

Selective Improvement

+2.1pp▲ Spatial	+1.3pp▲ Goal
+0.1pp – Object	-0.3pp – Long

Improves where uncertain, preserves where confident



High uncertainty -> CRM active

Gating correctly modulates intervention intensity



Low uncertainty -> CRM bypassed

Model Size



~327K

parameters (1.3 MB)

Latency



~170ms

all *K* samples batched

Best Improvement



+2.1pp

Spatial (~20% trigger)

Validated



SmolVLA

0.5B flow-matching VLA